EXPERIMENT NO. 7 WINDOWS SECURITY CONFIGURATION AND ACCESS AUDIT

AIM:

To audit Windows system configuration, Group Policy settings, user accounts, and access privileges using Windows tools and Sysinternals AccessChk to assess system security.

OBJECTIVES:

1. To examine Windows system configuration and operating system details to understand the current system environment.
2. To identify local user accounts and verify their group memberships and administrative privileges.
3. To audit applied Group Policy settings using gpresult and understand the available policy configurations.
4. To check PowerShell execution policies and identify settings that affect script execution and system security.
5. To inspect user directory permissions using Sysinternals AccessChk and verify the access rights assigned to the user.

HARDWARE & SOFTWARE REQUIREMENTS:

Category	Specification & Tools
Hardware Configuration	PC/Laptop with Intel Core i5 or higher, minimum 4 GB RAM, Internet Connection
Operating System	Windows 10/11
Network	Local system environment; internet required to download AccessChk
User Account Tools	Command Prompt and Windows PowerShell
System Audit Tools	winver, whoami, net user, gpresult
Security Policy Tools	PowerShell Execution Policy and Group Policy tools
Access Auditing Tool	Sysinternals AccessChk v6.15
Documentation Tools	Microsoft Word

THEORY:

Windows system auditing is the process of examining system configuration, user accounts, security policies, and access permissions to evaluate the security of a computer. It helps identify incorrect configurations, excessive privileges, and potential security risks that may affect the system.

Windows system configuration provides important information about the operating system, version, and build number. Tools such as winver and PowerShell commands are used to view these details and understand the current system environment. This information helps verify the operating system being audited.

User account auditing involves identifying local user accounts, checking group memberships, and verifying administrative privileges. Commands such as whoami, whoami /groups, net user, and net localgroup Administrators help examine user identities and their assigned permissions. This process helps determine which accounts have administrative access and whether privileges are appropriately assigned.

Group Policy is a Windows feature used to manage and control user and computer settings. The gpresult command displays the Group Policy settings applied to a particular user or computer. It helps verify whether policies are configured and applied successfully. However, some Group Policy management tools, such as secpol.msc and gpedit.msc, may not be available by default in Windows Home editions.

PowerShell execution policy controls the conditions under which PowerShell scripts can run. The Get-ExecutionPolicy -List command displays execution policy settings at different scopes, including the current user and local computer. Auditing these settings helps understand script execution restrictions and identify configurations that may require further review.

Access permissions determine whether users can read, write, modify, or execute files and folders. Sysinternals AccessChk is a security auditing tool that examines the permissions assigned to users and groups. In this practical, AccessChk is used to inspect the access rights of the current user on their profile directory. This helps verify the user's permissions on important personal files and folders.

The results obtained from these commands are analyzed to understand the system's security configuration. For example, group membership information helps identify administrative privileges, while Group Policy results show the policies applied to the system. Directory permission checks help determine whether access rights are assigned appropriately.

The combination of Windows commands, PowerShell, gpresult, and Sysinternals AccessChk provides a systematic method for auditing system configuration, user privileges, Group Policy settings, and directory permissions. This practical helps develop an understanding of Windows security mechanisms and supports the identification of potential configuration and permission-related risks.

WINDOWS SECURITY CONFIGURATION AND ACCESS AUDIT – PHASES, TOOLS & COMMANDS:

1. **System Configuration Audit**

   In this phase, the basic configuration of the Windows operating system is examined to identify the installed Windows edition, version, and build number. This information helps understand the system environment and verify the operating system on which the security audit is performed. The winver utility and PowerShell commands are used to collect the required system details.

Tool / Utility	Purpose	Command Used
Windows Run	View Windows version	winver
PowerShell	Check operating system details	Get-CimInstance Win32_OperatingSystem

2. **User Account and Privilege Audit**

   In this phase, local user accounts and their group memberships are examined to understand how user access is managed in Windows. The audit identifies the currently logged-in user, lists available local accounts, and checks which users belong to the Administrators group. These details help determine the privileges assigned to users and identify accounts with administrative access.

Tool / Utility	Purpose	Command Used
Command Prompt	Identify current user	whoami
Command Prompt	View user group memberships	whoami /groups
Command Prompt	List local user accounts	net user
Command Prompt	Check administrator group members	net localgroup Administrators

3. **Group Policy Audit**

   In this phase, Group Policy settings are examined to determine which policies are applied to the current user and computer. Group Policy helps manage security settings, user permissions, and system configurations. The gpresult command displays the available policy information and helps identify whether policies have been applied. The Group Policy Editor can also be checked if it is supported by the installed Windows edition.

Tool / Utility	Purpose	Command Used
Command Prompt	View applied user policies	gpresult /r
Command Prompt	Check computer-level policies	gpresult /scope computer /r
Run	Open Group Policy Editor, if supported	gpedit.msc

4. **PowerShell Execution Policy Audit**

In this phase, PowerShell execution policy settings are checked to understand the restrictions applied to PowerShell scripts. These policies help control script execution based on the configured security settings. The commands display the current execution policy and the policies configured at different scopes, such as the current user and local computer. The results help identify the settings that influence script execution in the system.

Tool / Utility	Purpose	Command Used
PowerShell	Check current execution policy	Get-ExecutionPolicy
PowerShell	View policies at all scopes	Get-ExecutionPolicy -List

5. **Network Connectivity Verification**

In this phase, file and directory access permissions are examined using Sysinternals AccessChk. The tool helps identify the permissions granted to a user on a specified directory. In this practical, the current user's profile directory is checked to determine the access rights available to that account. The results are reviewed to understand whether the user has read, write, or other access privileges on the specified location.

Tool / Utility	Purpose	Command Used
Sysinternals AccessChk	Check user directory permissions	accesschk.exe -d -v "%USERNAME%" "C:\Users\%USERNAME%"
File Explorer	View directory permissions	Properties → Security
AccessChk Output	Identify granted access rights	Review the displayed permissions

IMPLEMENTATION STEPS:

1. **System Configuration Audit**
   - Opened Windows Settings and examined the installed Windows edition and system configuration.
   - Executed winver to identify the Windows version and build number.
   - Opened PowerShell and retrieved operating system details using Get-CimInstance Win32_OperatingSystem.
   - Recorded the system information required for the security audit.
   - Verified the Windows environment before proceeding with further auditing steps.

2. **User Account and Privilege Audit**
   - Opened Command Prompt and executed whoami to identify the currently logged-in user.
   - Used whoami /groups to examine the user's group memberships and privileges.
   - Executed net user to list the local user accounts available on the system.
   - Used net localgroup Administrators to identify the members of the local Administrators group.
   - Reviewed the command outputs to understand user identities and assigned administrative privileges.

3. **Group Policy Audit**
   - Opened Command Prompt to examine the Group Policy settings applied to the system.
   - Executed gpresult /r to display the applied user-level Group Policy information.
   - Used gpresult /scope computer /r to check computer-level Group Policy information.
   - Examined the results to determine whether Group Policy settings were applied.
   - Checked the availability of gpedit.msc and secpol.msc and recorded the limitations encountered in Windows 11 Home Single Language.

4. **PowerShell Execution Policy Audit**
   - Opened Windows PowerShell to examine the script execution policy settings.
   - Executed Get-ExecutionPolicy to identify the effective PowerShell execution policy.
   - Used Get-ExecutionPolicy -List to display policies configured at different scopes.
   - Recorded the execution policy values for the current user, local computer, and other available scopes.
   - Analyzed the output to understand the restrictions affecting PowerShell script execution.

5. **Network Connectivity Verification**
   - Opened the folder containing the downloaded Sysinternals AccessChk utility.
   - Launched Command Prompt or PowerShell in the AccessChk directory.
   - Executed the following command to inspect the current user's profile directory permissions:

.\accesschk.exe -d -v "$env:USERNAME" "C:\Users\$env:USERNAME"

- Examined the output to identify the access rights granted to the user on the specified directory.
- Recorded the observed permissions and reviewed the results as part of the Windows security audit.

OBSERVATIONS:

During the cybersecurity practical, Kali Linux and Metasploitable2 were deployed on an isolated 192.168.56.0/24 Host-Only network. Nmap and Nikto were used to analyze the target, followed by controlled vulnerability testing using Metasploit.

- The Nmap scan successfully identified multiple services including vsftpd 2.3.4, OpenSSH, Telnet, SMTP, Apache, MySQL, PostgreSQL, VNC, and IRC, along with an outdated Linux kernel.
- The Nikto scan identified an outdated Apache/PHP installation, missing security headers, directory indexing, exposed phpinfo.php, and an accessible phpMyAdmin change log.
- The vsftpd 2.3.4 service was identified as a known vulnerable version, and the corresponding vsftpd backdoor exploit module was found in Metasploit.
- The Metasploit module was configured with RHOSTS 192.168.56.103 and LHOST 192.168.56.102 after identifying the Kali Linux IP address.
- The initial exploitation attempt failed because LHOST was not configured, but the error was resolved and the module was later fully configured; a successful callback or session was not confirmed in the provided output.

OUTPUT:

Phase 1 – System Configuration Audit



Fig. 1: Windows System Configuration

Phase 2 – User Account and Privilege Audit

```
C:\Users\salga>whoami
pooh\salga

C:\Users\salga>whoami /groups

GROUP INFORMATION
-----------------

Group Name                                                 Type             SID                                                
```

Phase 4 – PowerShell Execution Policy Audit

```
PS C:\WINDOWS\system32> Get-CimInstance Win32_OperatingSystem | Select-Object Caption, Version, BuildNumber

Caption                                  Version    BuildNumber
-------                                  -------    -----------
Microsoft Windows 11 Home Single Language 10.0.26200 26200


PS C:\WINDOWS\system32> Get-ExecutionPolicy -List

        Scope ExecutionPolicy
        ----- ---------------
MachinePolicy       Undefined
   UserPolicy       Undefined
      Process       Undefined
  CurrentUser    RemoteSigned
 LocalMachine       Undefined
```

Fig. 4: Auditing User-Level Group Policy Settings

Phase 5 – Network Connectivity Verification

```
C:\Users\salga>gpresult /r



                                                                                Waiting for other instances to complete their method execution
Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on 19-10-2026 at 19:26:40


RSOP data for POOH\salga on POOH : Logging Mode
-----------------------------------------------

OS Configuration:            Standalone Workstation
OS Version:                  10.0.26200
Site Name:                   N/A
Roaming Profile:             N/A
Local Profile:               C:\Users\salga
Connected over a slow link?: No


USER SETTINGS
--------------

    Last time Group Policy was applied: 19-10-2026 at 12:55:41
    Group Policy was applied from:      N/A
    Group Policy slow link threshold:   500 kbps
    Domain Name:                        POOH
    Domain Type:                        <Local Computer>

    Applied Group Policy Objects
    ----------------------------
        N/A

    The following GPOs were not applied because they were filtered out
    -------------------------------------------------------------------
        Local Group Policy
            Filtering:  Not Applied (Empty)

    The user is a part of the following security groups
    ---------------------------------------------------
        High Mandatory Level
        Everyone
        Local account and member of Administrators group
        BUILTIN\Administrators
        BUILTIN\Users
        Performance Log Users
        NT AUTHORITY\INTERACTIVE
        CONSOLE LOGON
        NT AUTHORITY\Authenticated Users
        This Organization
        salgaonkarpurva@icloud.com
        Local account
        LOCAL
        Cloud Account Authentication
```

Fig. 5: Auditing Computer-Level Group Policy Settings

```
C:\Users\salga\Downloads\AccessChk>accesschk.exe -d -v "%USERNAME%" "C:\Users\%USERNAME%"

Accesschk v6.15 - Reports effective permissions for securable objects
Copyright (C) 2006-2022 Mark Russinovich
Sysinternals - www.sysinternals.com

RW C:\Users\salga
        FILE_ALL_ACCESS
```

Fig. 6: Auditing User Directory Permissions Using AccessChk

ANALYSIS:

The experiment successfully demonstrated the auditing of Windows system configuration, user accounts, Group Policy settings, PowerShell execution policies, and directory permissions. Windows commands and Sysinternals AccessChk were used to examine system information and user privileges. Some limitations were encountered, including the unavailability of secpol.msc and access denial during the computer-level Group Policy audit.

PREVENTIVE MEASURES:

1. Keep the Windows operating system and installed applications updated with the latest security patches.
2. Assign user accounts only the permissions required to perform their tasks.
3. Review administrator group memberships regularly to prevent unnecessary privileged access.
4. Configure appropriate PowerShell execution policies and review security settings periodically.
5. Audit Group Policy settings and file permissions regularly to identify and correct potential security weaknesses.

CONCLUSION:

The experiment successfully demonstrated the auditing of Windows system configuration, user accounts, Group Policy settings, and access privileges using built-in Windows tools and Sysinternals AccessChk. The practical helped identify system security settings, examine user permissions, and understand the limitations of the Windows Home edition.